Theorem 7.8 (Extreme Value Theorem)

If $f(x)$ is continuous on a closed interval $[a,b]$, then f has both an absolute maximum and an absolute minimum on $[a,b]$.

The absolute extrema of $f(x)$ occur either at the endpoints of closed interval $[a,b]$ or inside the open interval (a,b).If the absolute extrema occurs inside, then it must occur at critical numbers of $f(x)$. Thus, we can use the following procedure to find the absolute extrema of a continuous function on closed interval $[a,b]$.

A procedure for finding the absolute extrema of a continuous function $f(x)$ on closed interval $[a,b]$.

Step 1 : Find the critical numbers of $f(x)$ in (a,b)

Step 2 : Evaluate $f(x)$ at all the critical numbers and at the endpoints a and b

Step 3 : The largest and the smallest of the values in step 2 is the absolute maximum and absolute minimum of $f(x)$ respectively on the closed interval $[a,b]$.

Example 7.48

Find the absolute maximum and absolute minimum values of the function $f(x)=2x^3+3x^2-12x$ on $[-3,2]$

Solution

Differentiating the given function, we get

$$\begin{aligned} f'(x) &= 6x^2+6x-12 \\ &= 6(x^2+x-2) \\ f'(x) &= 6(x+2)(x-1) \end{aligned}$$

Thus, $f'(x)=0 \Rightarrow x=-2,1 \in (-3,2)$.

Therefore, the critical numbers are $x=-2,1$. Evaluating $f(x)$ at the endpoints $x=-3,2$ and at critical numbers $x=-2,1$, we get $f(-3)=9$, $f(2)=4$, $f(-2)=20$ and $f(1)=-7$.

From these values, the absolute maximum is 20 which occurs at $x=-2$, and the absolute minimum is -7 which occurs at $x=1$. ■

Example 7.49

Find the absolute extrema of the function $f(x)=3\cos x$ on the closed interval $[0,2\pi]$.

Solution

Differentiating the given function, we get $f'(x)=-3\sin x$.

Thus, $f'(x)=0 \Rightarrow \sin x=0 \Rightarrow x=\pi \in (0,2\pi)$. Evaluating $f(x)$ at the endpoints $x=0, 2\pi$ and at critical number $x=\pi$, we get $f(0)=3$, $f(2\pi)=3$, and $f(\pi)=-3$.

From these values, the absolute maximum is 3 which occurs at $x=0,\ 2\pi$, and the absolute minimum is -3 which occurs at $x=\pi$. ■

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